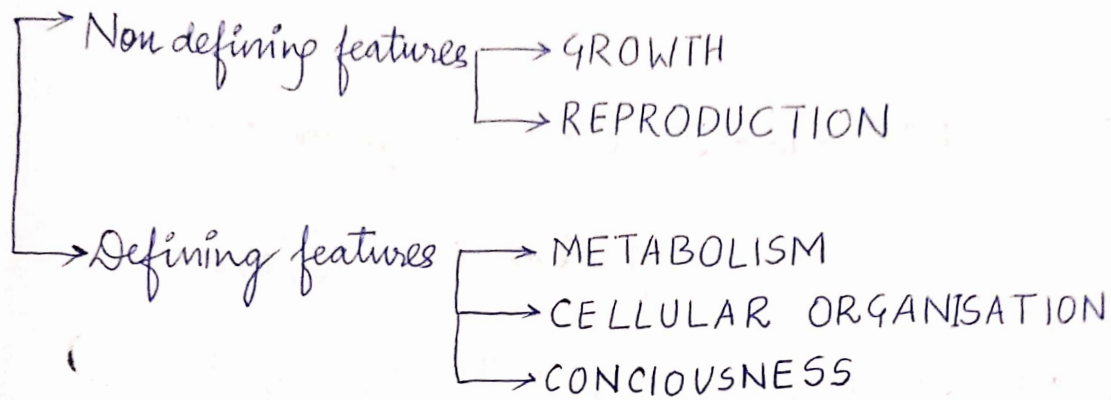


THE LIVING WORLD



→ What is living :- features that are defining and non-defining



① GROWTH:- Increase in mass and number of organisms

- Multicellular - grow by cell division.
- Unicellular - grow by cell division

Exception - non-living such as mountain, boulder, sand mound do grow

∴ Growth cannot be taken as a defining feature

② REPRODUCTION:- in multicellular organisms - reproduction refers to the production of progeny possessing features more or less similar to those of parents.

- Fungi produce spores to multiply.
- Yeast and hydra → Budding
- Planaria - True regeneration
- Filamentous algae - Fragmentation.

→ In unicellular organisms - growth and reproduction is synonymous with growth - i.e. increase in number of cells.

→ Many organisms which do not reproduce (mules, sterile, worker bees, infertile human couples.

∴ Reproduction also cannot be an all-inclusive defining characteristic



→ ③ METABOLISM :- The sum total of all types of chemical reactions occurring in an individual due to specific interaction.

→ No non-living object exhibits metabolism.

→ Hence, metabolism is a defining feature.

→ ④ CELLULAR ORGANISATION :-

→ MULTICELLULAR :- Cell → Tissue → Tissue System → Organ

Organism ← Organ System

→ Cellular organisation of the body is the defining feature of life forms.

→ ⑤ CONSCIOUSNESS :-

→ It is the ability to sense their surrounding or environment and respond to these environmental stimuli which could be physical, chemical and biological.

→ Human beings is the only organism who is aware of of himself → has self consciousness.

∴ Consciousness is therefore defining features of life forms.

→ DIVERSITY IN THE LIVING WORLD :-

→ BIODIVERSITY :- The variety of life on Earth.

→ Number of species that are known and described between 1.7-1.8 million.

→ Since organisms have name that vary from place to place. Thus to recognise the organisms we are referring to, a standard system is needed.

→ Organisations:- provide scientific names based on principles and criteria

- ① ICBN - International Code of Botanical Nomenclature
- ② ICZN - International Code of Zoological Nomenclature
- ③ ICBN - International Code of Bacteriological Nomenclature
- ④ ICVN - International Code of Viral Nomenclature
- ⑤ ICNCP - International Code of Nomenclature for Cultivated Plants.

→ TAXONOMY:- This is the process of classification of different living organism into different categories is taxonomy

→ Steps:- C I C N

- ① Characterisation
- ② Identification
- ③ Classification
- ④ Nomenclature

→ The scientific term for these taxonomic categories is Taxon/Taxa

→ All taxonomic categories together constitute taxonomic hierarchy.

→ SYSTEMATICS → Word Systematics is derived from Latin word 'systema' → systematic arrangement of organisms.

→ It tells about the different kind of organisms, their diversities and also the relationships (evolutionary relationship) among them.

→ Systematics → Taxonomy + Evolutionary Relationships (phylogenetic)

Term was given by { Carlous Linnaeus.
Book - Systema Nature.

